PRESIDENCY UNIVERSITY
Private University Estd. in Karnataka State by Act No. 41 of 2013
Itgalpura, Rajankunte, Yelahanka, Bengaluru – 560064

Development of a Digital Farm Management Portal for implementing Biosecurity measures in Pig and Poultry Farms

A PROJECT REPORT

Submitted by

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IN

COMPUTER SCIENCE AND ENGINEERING

PRESIDENCY UNIVERSITY

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PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

BONAFIDE CERTIFICATE

Certified that this report **"Development of a Digital Farm Management Portal for implementing Biosecurity measures in Pig and Poultry Farms"** is a bonafide work of "YANAMALA JASWANTH REDDY(20221CSE0454), PUROHITH SATYANARAYANA (20221CSE0451), AMAREGOUDA(20221CSE0506)", who have successfully carried out the project work and submitted the report for partial fulfilment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE AND ENGINEERING during 2025-26.

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DECLARATION

We the students of final year B. Tech in COMPUTER SCIENCE AND ENGINEERING at Presidency University, Bengaluru, named YANAMALA JASWANTH REDDY, PUROHITH SATYANARAYANA, AMAREGOUDA, hereby declare that the project work titled **"Development of a Digital Farm Management Portal for implementing Biosecurity measures in Pig and Poultry Farms"** has been independently carried out by us and submitted in partial fulfillment for the award of the degree of B. Tech in COMPUTER SCIENCE AND ENGINEERING during the academic year of 2025-26. Further, the matter embodied in the project has not been submitted previously by anybody for the award of any Degree or Diploma to any other institution.

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Abstract

The pig and poultry industries have become victims of fragmented and manual record keeping in farm management, particularly in small and medium farms. Critical information regarding animal health, vaccination consultations, feed controls, environmental factors, and biosecurity controls is normally dispersed through registers or individual systems. This renders it difficult to have a clear picture of what is happening to the farmers and administrators. Inefficiencies, slowness in detection of diseases and challenges in adhering to the standard protocols of biosecurity are the results of a lack of integration. Although there are some digital solutions, they are generally too generic, fail to provide real-time monitoring, and may not also directly address the biosecurity requirements of livestock farming. The Digital Farm Management Portal is meant to address this issue by offering a centralized, interactive and easy to use site. It incorporates the necessary information on the farms and lays a lot of emphasis to the biosecurity on the pig and poultry farms. The portal combines over 4 Key Performance Indicators (KPIs) in areas that are very important like animal health, vaccination, feed control, Livestock Tracking, hygiene practices and disease control. The system is scalable, transparent, and cost-effective using modern open-source technologies, including React.js on the frontend, Flask or Django on the backend, MySQL on the data storage side, and Pandas on the data processing and visualization side. The installation can be used with real-time updates of data through automated ETL (Extract, Transform, Load) pipelines. It helps farmers and administrators to monitor the situation of the farms and biosecurity compliance using accurate and up-to-date data. The system has essential features such as real-time monitoring, interactive visualization, and automated generation of reports in PDF and Excel format to manage the farms effectively. The portal has role-based operation meaning that it is used securely by farm owners, veterinarians, and farm workers. Such functions as vaccine monitoring, health monitoring, biosecurity compliance check, notification of alerts, and analysis of environmental conditions assist users in managing farms. Moreover, data-driven insights and trend analysis help to detect diseases early, enhance productivity and ensure high levels of biosecurity in pig and poultry farms.The solution covers the gap of the traditional way of managing the farm. The need for a combined digital system exists.